Sprint Report

REPORT 2

Unit code: EAT40005
Unit Name: Engineering Project A
Submission date: 17/05/2026

Student Name	Student Id	Statement of contribution to the report
Rumeth Gunasinghe	**104498227**	**Sprint Plan, Progress, Demonstation**
Thinidu randin	**104675729**	**Sprint Review, Retrocspect, lessons learned**

1. ACKNOWLEDGMENT OF COUNTRY

We acknowledge the Traditional Owners of the lands on which this work was completed. We recognise the Wurundjeri Woi Wurrung people of the Kulin Nation as the custodians of the land in Melbourne, Australia, where this project work was undertaken. We pay our respects to their Elders past, present, and emerging.
We acknowledge their continuing connection to land, waters, and culture, and recognise the important role that Indigenous knowledge and perspectives play in shaping sustainable and respectful technological development.
As team members working and studying in Australia, we recognise and respect the cultural heritage of the Traditional Owners of the lands on which we live and learn.

2. SPRINT PLAN

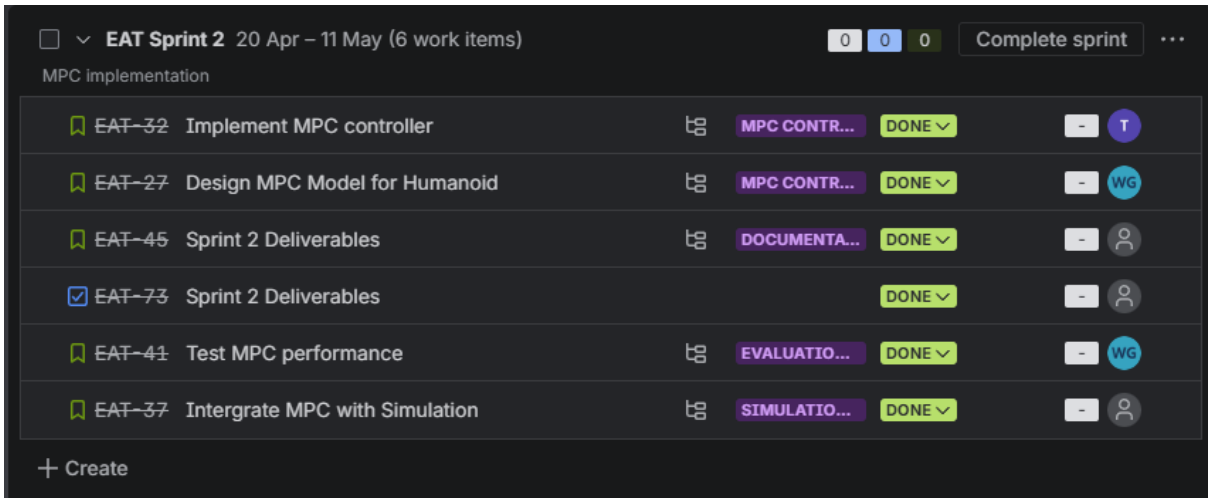


Figure 1: Sprint 2 Jira Board and Sprint Planning showing task allocations and sprint tracking.

Sprint 2 focused on extending the foundational control concepts developed during Sprint 1 into a more realistic humanoid robotics environment. While Sprint 1 primarily concentrated on balancing an inverted pendulum system using PID and MPC control approaches, Sprint 2 aimed to transition these concepts toward humanoid robot balance control using a full-body simulation framework.

The primary objective of this sprint was to establish a stable humanoid simulation environment capable of maintaining temporary balance using low-level stabilization techniques. To achieve this, the sprint focused on migrating from the PyBullet physics engine to MuJoCo due to its improved physics realism and suitability for articulated humanoid robotics simulations.

The sprint also aimed to integrate the Unitree H1 humanoid model into the MuJoCo environment and implement balancing strategies including PID-based stabilization, MPC-inspired predictive balancing, and center of mass (COM) estimation. Additional goals included improving posture stability through crouched stance optimization, locking unnecessary degrees of freedom, and tuning joint damping and actuator gains.

The deliverables planned for Sprint 2 included:

1. A functioning MuJoCo humanoid simulation environment
2. Successful integration of the Unitree H1 humanoid model
3. Functional PID stabilization controller
4. Initial MPC-inspired balancing implementation
5. COM estimation and posture stabilization
6. Temporary humanoid balance demonstration
7. Structured sprint documentation and reporting

3. SPRINT PROGRESS

Sprint 2 focused on implementing and stabilizing a humanoid robot balancing framework using the MuJoCo physics engine and the Unitree H1 humanoid model. The sprint goal was to successfully migrate from the previous PyBullet environment and establish a realistic simulation architecture capable of maintaining temporary humanoid balance.

Work	Assignee	Reporter	Priority	Status	Resolution	Created	Updated
~~EAT-71~~ Implement MPC	Weligama Arachchig...	Weligama Arach...	Medium	DONE	Done	Apr 19, 2026, 7:17 PM	Apr 19, 2026, 7:21 PM
~~EAT-49~~ Collect Client feedback	Weligama Arachchig...	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:11 PM	May 17, 2026, 8:16 PM
~~EAT-46~~ Prepare Demonstration	Weligama Arachchig...	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:11 PM	May 17, 2026, 8:16 PM
~~EAT-44~~ Record performance data	Weligama Arachchig...	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:10 PM	May 17, 2026, 8:17 PM
~~EAT-41~~ Test MPC performance	Weligama Arachchig...	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:10 PM	May 17, 2026, 8:15 PM
~~EAT-39~~ Run closed loop simulation	Weligama Arachchig...	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:09 PM	May 17, 2026, 8:17 PM
~~EAT-38~~ Connect MPC output to system input	Weligama Arachchig...	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:09 PM	May 17, 2026, 8:17 PM
~~EAT-36~~ Generate control output	Weligama Arachchig...	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:08 PM	May 17, 2026, 8:17 PM
~~EAT-34~~ Implement prediction Model	Weligama Arachchig...	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:08 PM	May 17, 2026, 8:17 PM
~~EAT-33~~ Setup Optimisation framework	Weligama Arachchig...	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:08 PM	May 17, 2026, 8:17 PM
~~EAT-30~~ Define cost function	Weligama Arachchig...	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:07 PM	May 17, 2026, 8:16 PM
~~EAT-28~~ Define system state	Weligama Arachchig...	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:07 PM	May 17, 2026, 8:16 PM
~~EAT-27~~ Design MPC Model for Humanoid	Weligama Arachchig...	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 3:05 PM	May 08, 2026, 9:28 AM
~~EAT-26~~ Write sprint report - part II	Weligama Arachchig...	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 2:56 PM	Apr 24, 2026, 9:37 AM

Figure 2: Tasks assigned to Rumeth in the Jira Board during Sprint 2.

From a development perspective, Rumeth Gunasinghe primarily focused on the migration from PyBullet to MuJoCo, integration of the Unitree H1 MJCF model, implementation of balancing controllers, and stabilization tuning. This included debugging actuator mappings, implementing COM estimation logic, tuning joint torques, and designing the crouched balancing posture used throughout the simulation.

Work	Assignee	Reporter	Priority	Status	Resolution	Created	Updated
~~EAT-72~~ PID testing and logging	Thinidu	Weligama Arach...	Medium	DONE	Done	Apr 19, 2026, 7:43 PM	Apr 19, 2026, 7:44 PM
~~EAT-48~~ Write sprint report	Thinidu	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:11 PM	May 17, 2026, 8:16 PM
~~EAT-47~~ Ducument MPC implementation	Thinidu	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:11 PM	May 17, 2026, 8:16 PM
~~EAT-43~~ Compare behavior with PID	Thinidu	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:10 PM	May 17, 2026, 8:17 PM
~~EAT-42~~ Run test scenarios	Thinidu	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:10 PM	May 17, 2026, 8:17 PM
~~EAT-40~~ Debug instability issues	Thinidu	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:09 PM	May 17, 2026, 8:17 PM
~~EAT-35~~ Implement optimisation Model	Thinidu	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:08 PM	May 17, 2026, 8:17 PM
~~EAT-32~~ Implement MPC controller	Thinidu	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:07 PM	May 08, 2026, 9:27 AM
~~EAT-31~~ Document MPC formulation	Thinidu	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:07 PM	May 17, 2026, 8:16 PM
~~EAT-29~~ Define prediction horizon	Thinidu	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 5:07 PM	May 17, 2026, 8:16 PM
~~EAT-25~~ Write sprint report	Thinidu	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 2:56 PM	Apr 24, 2026, 9:37 AM
~~EAT-15~~ Implement PID Controller	Thinidu	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 2:49 PM	Apr 19, 2026, 7:20 PM
~~EAT-11~~ Research Control Methods	Thinidu	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 2:43 PM	Apr 19, 2026, 7:21 PM
~~EAT-6~~ Documentation and Report	Thinidu	Weligama Arach...	Medium	DONE	Done	Apr 10, 2026, 2:29 PM	Apr 19, 2026, 7:44 PM

Figure 3: Tasks assigned to Thinidu in the Jira Board during Sprint 2.

Thinidu Madurapperuma contributed to controller refinement, debugging joint behaviour, improving actuator responsiveness, and assisting with simulation testing and documentation. Thinidu also contributed toward analysing humanoid posture instability and reviewing controller behaviour under balancing disturbances.

A major milestone during this sprint was the successful migration from PyBullet to MuJoCo. Several limitations were encountered in the original PyBullet environment, including unrealistic humanoid contact behaviour, unstable balancing dynamics, and poor actuator

responsiveness. These limitations significantly affected humanoid balancing performance and motivated the transition toward MuJoCo.

The Unitree H1 humanoid model was successfully loaded into the MuJoCo simulation using MJCF scene integration. Joint mapping and actuator indexing were implemented to enable low-level torque-based control of hip, knee, and ankle joints.

The sprint deliverables achieved include:

1. Successful migration from PyBullet to MuJoCo
2. Integration of the Unitree H1 humanoid robot model
3. Implementation of PID stabilization control
4. Development of MPC-inspired balancing logic
5. COM estimation module implementation
6. Joint locking and posture stabilization
7. Temporary humanoid balancing demonstration lasting approximately 4–5 seconds



Figure 4: Current project repository showing latest commits and updated MuJoCo humanoid balancing framework pushed by Rumeth and Thinidu.

4. SPRINT DEMONSTRATION

The Sprint 2 demonstration session was conducted on 16/05/2026 with the participation of both team members and the client representative. The meeting was held online via Microsoft Teams.

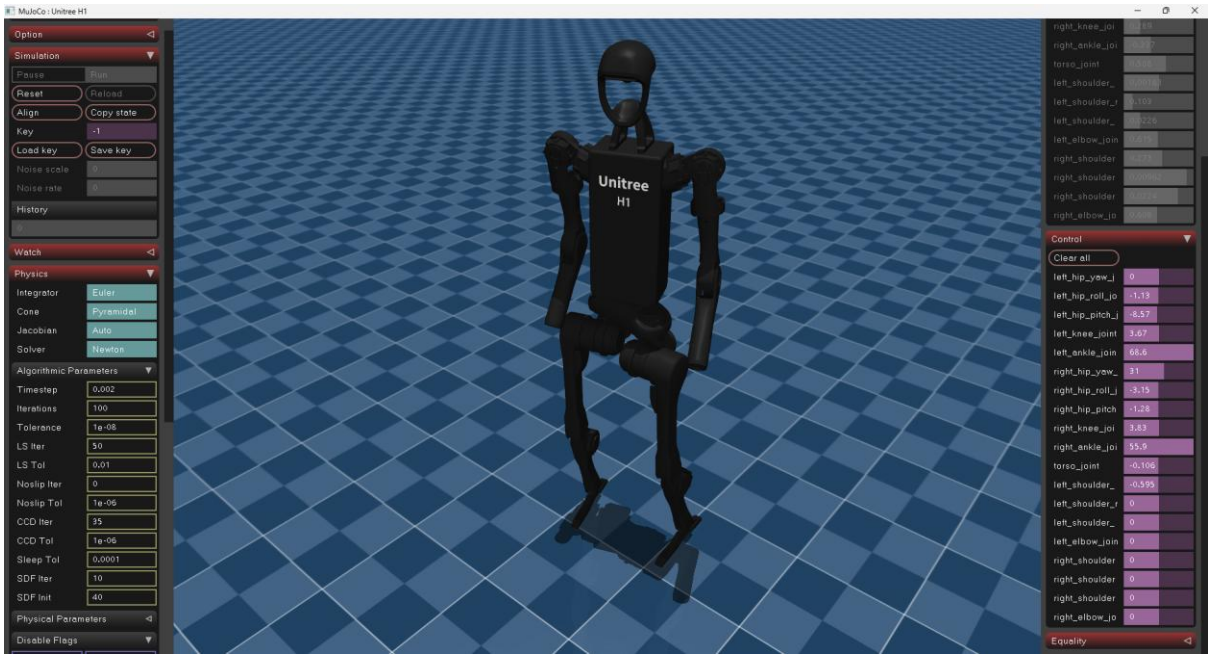


Figure 6: MuJoCo humanoid simulation demonstrating crouched stabilization posture.

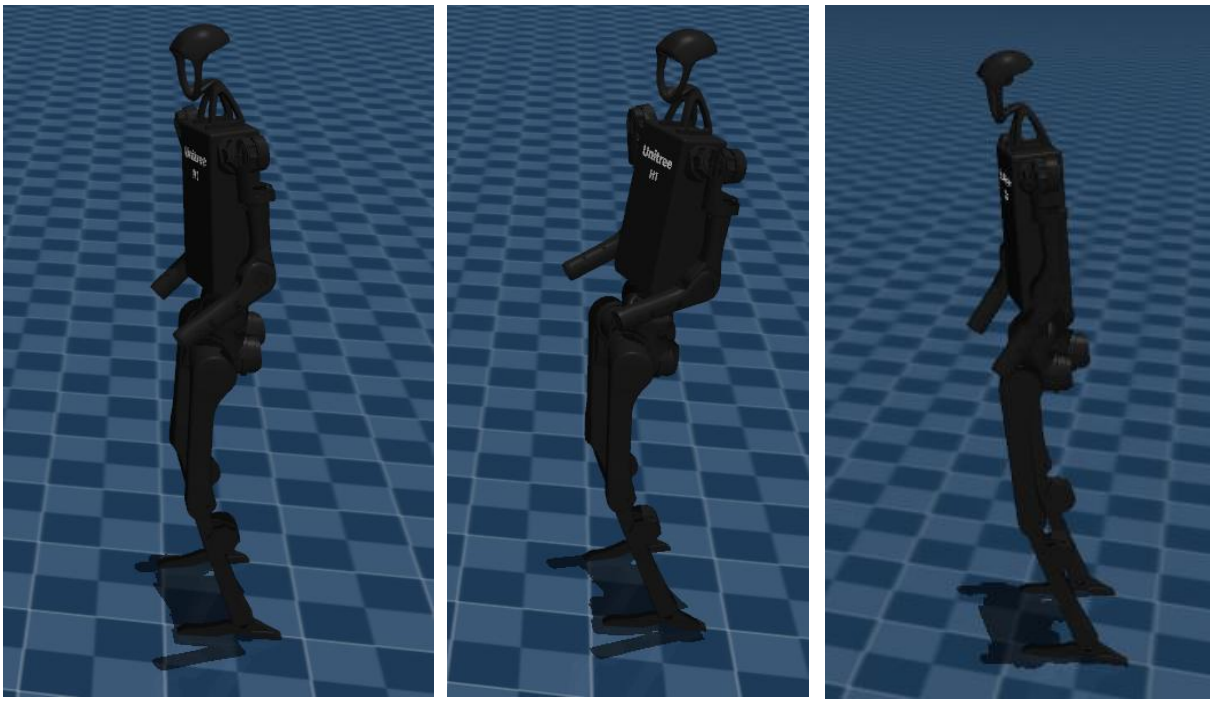


Figure 5 : Active balancing behaviour of the Unitree H1 humanoid during simulation.

The objective of the demonstration was to present the progress made toward humanoid robot balancing using the MuJoCo physics engine and the Unitree H1 humanoid robot model. The demonstration focused on explaining the transition from PyBullet to MuJoCo, the humanoid stabilization architecture, and the balancing behaviour achieved during simulation.

During the demonstration, the team explained the limitations previously encountered in PyBullet, including unrealistic balancing behaviour, unstable foot-ground interactions, and inaccurate humanoid dynamics. The reasons for selecting MuJoCo as the new simulation platform were discussed, particularly its improved articulated rigid-body dynamics and greater realism for humanoid robotics research.

The team demonstrated:

- MuJoCo simulation environment setup
- Unitree H1 humanoid integration
- PID stabilization controller
- MPC-inspired balancing controller
- COM estimation module
- Joint stabilization and crouched balancing posture

The demonstration showed the humanoid robot successfully entering a crouched posture, stabilizing temporarily, and actively correcting forward and backward instability before eventually falling after several seconds.

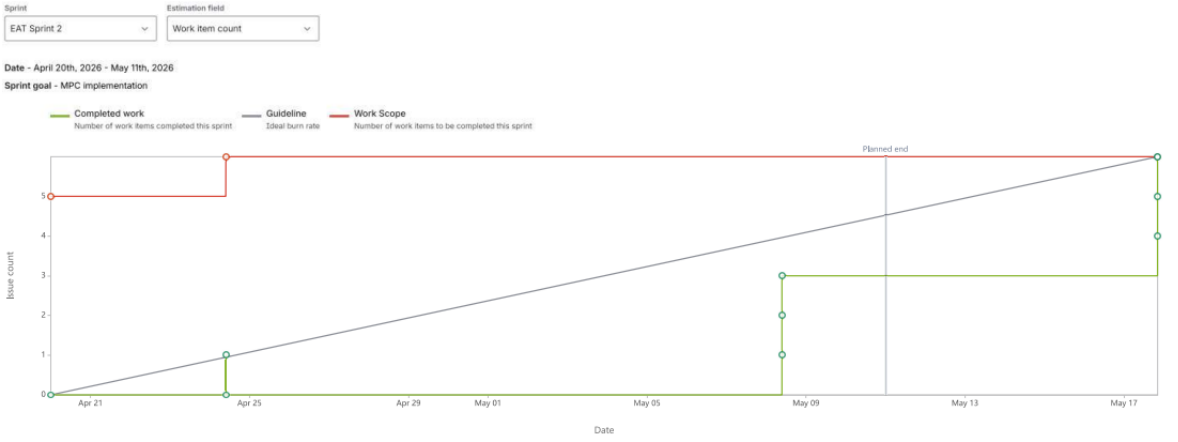
The client responded positively regarding:

- The migration to MuJoCo
- Improved realism of the humanoid simulation
- Modular controller architecture
- Progress made in humanoid balancing

The client acknowledged that humanoid balancing is a highly complex problem and recognised that temporary stabilization represented meaningful progress. Additional discussion focused on future work involving locomotion, stepping recovery, and more advanced predictive control systems.

As an outcome of the session, the client confirmed that the project was progressing in the correct direction and recommended focusing on improving recovery stabilization, whole-body balancing, and locomotion planning during the next sprint.

5. SPRINT REVIEW (CRITICAL REVIEW OF THE PRODUCT)



Date	Event	Work item	Completed	Scope
Mon, Apr 20 2026, 12:00am	Sprint started	EAT-27 Design MPC Model for Humanoid EAT-32 Implement MPC controller EAT-37 Intergrate MPC with Simulation EAT-41 Test MPC performance EAT-45 Sprint 2 Deliverables	0	5
Fri, Apr 24 2026, 9:38am	Work item completed	EAT-27 Design MPC Model for Humanoid	0 → 1	5
Fri, Apr 24 2026, 9:40am	Work item re-opened	EAT-27 Design MPC Model for Humanoid	1 → 0	5
Fri, Apr 24 2026, 9:43am	Added to sprint	EAT-73 Sprint 2 Deliverables	0	5 → 6
Fri, May 08 2026, 9:27am	Work item completed	EAT-32 Implement MPC controller	0 → 1	6
Fri, May 08 2026, 9:27am	Work item completed	EAT-37 Intergrate MPC with Simulation	1 → 2	6
Fri, May 08 2026, 9:28am	Work item completed	EAT-27 Design MPC Model for Humanoid	2 → 3	6
Sun, May 17 2026, 8:15pm	Work item completed	EAT-41 Test MPC performance	3 → 4	6
Sun, May 17 2026, 8:15pm	Work item completed	EAT-73 Sprint 2 Deliverables	4 → 5	6
Sun, May 17 2026, 8:15pm	Work item completed	EAT-45 Sprint 2 Deliverables	5 → 6	6
Sun, May 17 2026, 8:17pm	Sprint completed	EAT-27 Design MPC Model for Humanoid EAT-32 Implement MPC controller EAT-37 Intergrate MPC with Simulation EAT-41 Test MPC performance EAT-45 Sprint 2 Deliverables EAT-73 Sprint 2 Deliverables	6	6

The progress we got during Sprint 2 was evaluated against the begining sprint plan, which focused on establishing a realistic humanoid balancing framework using MuJoCo and the Unitree H1 humanoid robot model.

Overall, the sprint objectives was successfully achieved. The team successfully moved from PyBullet to MuJoCo, integrated the humanoid robot model, implementing balancing controllers, and demonstrated temporary humanoid stabilization behavior. These achievements show major progress compared to Sprint 1, which focused primarily on simplified cart-pole balancing systems.

One of the key accomplishments of this sprint was the development of a functioning humanoid stabilization framework using low level balancing control. From iterative tuning and debugging, the team achieved several seconds of active balancing behavior before instability accumulated and caused collapse.

The sprint also significantly improved our understanding of humanoid posture dynamics, center of mass alignment, and the complexity of articulated balance control systems. Major discoveries were made regarding crouched posture stability, pelvis orientation effects, ankle overcompensation, and the relationship between COM drift and falling behavior.

However, many challenges were encountered throughout. One of the key challenges involved joint mapping and actuator indexing within the Unitree H1 model. Incorrect actuator assumptions and mismatched joint indexing initially caused unstable behavior and invalid torque application.

Another main challenge involved balancing stability itself. Straight leg humanoid configurations proved highly unstable, requiring of implementation of a crouched posture to improve COM alignment and support geometry. Also, instability was caused by pelvis rotation, torso misalignment, and ankle overcorrection.

The client feedback gave during the demonstration showed that achieving temporary humanoid stabilization is already meaningful in progress within robotics research. The client also marked the importance of future improvements involving stepping recovery, locomotion planning, and advanced predictive balancing approaches.

From a reflective perspective, the sprint shows the importance of iterative development, incremental testing, and modular controller architecture design. Other than attempting full locomotion immediately, focusing on stabilization first allowed the team to progressively understand humanoid balancing dynamics in a controlled manner.

Future development will focus on:

- Recovery stepping
- Whole-body stabilization
- Advanced MPC implementation
- Reinforcement learning integration
- Improved state estimation systems

6. RETROSPECT (CRITICAL REVIEW OF THE PROCESS)

During Sprint 2, we generally followed the planned agile development process through continuous communication, iterative testing, sprint task tracking, and collaborative debugging. Jira was used for sprint tracking, task allocation, and monitoring development progress throughout the sprint.

One of the main process related challenges encountered during this sprint was the increased technical complexity of humanoid robotics simulation compared to the previous cart-pole system. Unlike simplified balancing models, the Unitree H1 humanoid introduced multiple interconnected joints, actuator mappings, and highly sensitive balance dynamics that required extensive debugging and tuning.

Another challenge involved the migration from PyBullet to MuJoCo. Although the transition mainly improved simulation realism, it introduced additional complexity related to MJCF integration, actuator indexing, and understanding MuJoCo specific simulation behavior. Considerable time was spent on troubleshooting low level simulation issues before meaningful balancing behaviour could be achieved.

Communication between team members remained consistent through Discord discussions, collaborative debugging sessions, and shared testing updates. Regular communication helped reduce duplicated work and ensured that all members remained aligned regarding controller changes and simulation adjustments.

From a quality assurance perspective, the sprint focused heavily on iterative simulation testing and stability evaluation. Therefore, more structured logging of tuning parameters and balancing metrics would improve future experimentation and reproducibility. Future sprints would benefit from automated parameter logging and more formalized testing procedures.

Risk management continued to play an important role throughout the sprint. Key risks identified included simulation instability, incorrect actuator mapping, balancing failure, excessive degrees of freedom, and unrealistic humanoid posture behavior. These risks were mitigated through incremental testing, posture simplification, joint locking, and controller tuning.

Ethically, the project remained entirely simulation based during Sprint 2, meaning no real world safety risks were introduced. so, the team continued to consider broader ethical implications surrounding humanoid robotics, particularly the importance of safe control systems, predictable behavior, and responsible autonomous robotics development.

Overall, Sprint 2 established a strong technical foundation for future humanoid locomotion research. However, stronger experimental logging, improved tuning methodologies, and more structured testing workflows would improve future sprint effectiveness.

7. LESSONS LEARNED (CRITICAL REVIEW OF SPRINT TWO EXPERIENCE AND FUTURE PLAN)

Lessons Learned

- Humanoid balancing is significantly more complex than simplified inverted pendulum systems.
- Accurate physics simulation is critical for realistic humanoid control behavior.
- The center of mass alignment is one of the most important factors affecting humanoid stability.
- Straight-leg humanoid configurations are less stable than crouched balancing postures.
- Incremental debugging and modular controller design greatly simplify complex robotics development.
- MuJoCo provides significantly more realistic articulated dynamics compared to PyBullet for humanoid robotics.
- Joint locking and DOF simplification are effective for reducing instability during early balancing development.
- Continuous simulation testing is essential when tuning humanoid control systems.

To-Do for Next Sprint

- Implement recovery stepping behavior for instability correction.
- Develop more advanced MPC balancing algorithms.
- Improve COM estimation and state prediction accuracy.
- Investigate reinforcement learning integration for adaptive balancing.
- Explore Isaac Sim compatibility for future scalability.
- Continue refining whole body balancing and locomotion behavior.